

Design and Development Report

Student Name: Showmen Sarker

Student ID: 52532619

Render Deployment URL:

<https://loan-analysis-application-flask-based.onrender.com>

GitHub Repository URL:

<https://github.com/showmensarker/Loan-Analysis-Application-Flask-Based>

Git branching and version control practices were also used during development to manage analytics improvements, testing updates, and interface enhancements before merging features into the main application.

Admin Login Credentials:

Username: admin

Password: admin123

The name of the project is “Loan Analysis Flask Application,” and it was built using Flask, a web framework based on the Python language. The application is a web-based tool intended to help manage and analyze data from customers and loans based on a loan default dataset.

In terms of the back-end development, SQLAlchemy serves as an ORM, making it easier to handle database transactions. Moreover, SQLite serves as an RDBMS, which facilitates the storage of customer and loan-related data. As for front-end development, HTML, CSS, and Jinja templates have been used.

The software application has all the capabilities for managing customers and their loans. The administrator is capable of viewing, adding, updating, and deleting customers’ and loans’ information from the software application. Customers’ loan history information can be viewed from the software application. Pagination capability was added in the customer and loans section of the software application to facilitate easy interaction with larger amounts of data.

Analysis dashboard was created to generate statistical data. This included the average income for the customer, average loan amount, the highest loan amount issued, total number of defaulted loans, and the number of high-risk customers. Two charts were designed using Matplotlib for visualizing the loan risk and loan defaults per age group. The dashboard interface was further enhanced with responsive mobile-friendly design improvements, optimized chart scaling for larger datasets, risk categorization cards, and additional analytical insights to improve usability and data visualization.

The system also incorporates advanced filtering and sorting capability, where customers can be searched by customer ID and sorted according to highest or lowest income levels. Loans can be filtered according to minimum credit score, loan risk category, and customer ID search functionality. Custom HTTP error pages have been developed for codes 400, 403, 404, 405, and 500.

With regards to security, administrator authentication has been incorporated with the use of Flask’s session management tool. Login credentials are needed in order to perform CRUD actions, and there is a feature that allows users to log out of the application.

Testing was done through automation using Pytest in order to make the application more reliable and correct. The tests included functional testing, route testing, authentication testing, pagination testing, validation testing, filtering and sorting tests, edge case testing, and HTTP error handling tests. Pytest fixtures were also used to make tests easier to manage and above 60 automated tests were executed successfully without issues throughout the application.

Lastly, the Loan Analysis Flask Application demonstrates the development of a secure and well-structured web application that can manage and analyze customer and loan data effectively. The project focuses on data analysis, visualization, authentication, CRUD functionality, and automated testing features to create a complete and simple user-friendly system.